An Internship Report

Of

Web Development

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student:- 25SCS1003000810

Name of Student: VAIBHAV GUPTA

Batch: 2025-2029

CANDIDATE'S DECLARATION

I, VAIBHAV GUPTA, do hereby declare that the internship report titled "Web Development" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: VAIBHAV GUPTA

Roll No.: 25SCS1003000810

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Prodigy InfoTech for offering me the opportunity to work as a Web Development Intern and for their guidance and support throughout the internship period. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: VAIBHAV GUPTA

Roll No.: 25SCS1003000810

TABLE OF CONTENTS

3. INTERNSHIP COMPLETION CERTIFICATE

PRODIGY INFOTECH

CERTIFICATE

OF COMPLETION

CIN: PIT/AUG26/00972

Issued on: 01/09/2026

This Certificate is proudly presented to

Vaibhav Gupta

for successfully completing a **1-month** internship from **1st August, 2026** to **31st August, 2026** in **Web Development** with outstanding remarks at Prodigy InfoTech.

contact@prodigyinfotech.dev

www.prodigyinfotech.dev

4. PROJECT DESCRIPTION

1. Introduction

This report presents a detailed account of the 1-month internship undertaken as a Web Development Intern at Prodigy InfoTech, carried out from 1st August 2026 to 31st August 2026. The internship was completed as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme, 3rd Semester, at IILM University, Greater Noida. The primary aim of the internship was to strengthen practical, front-end web development skills by building real-world, browser-based applications using HTML, CSS, and JavaScript.

Over the course of the internship, three hands-on tasks were assigned and completed, each reinforcing a different aspect of client-side web development — page structuring and styling, DOM manipulation and timing logic, and interactive, event-driven game design. On successful completion of the internship, a Certificate of Completion was awarded by Prodigy InfoTech with outstanding remarks.

2. Organization Profile

Prodigy InfoTech is an MSME-registered organization (registered under Udyam, Udyam-MH-19-0221586) that offers structured, educational internship programs to students across various technology domains. The organization is identified under Company Identification Number PIT/AUG26/00977 and provides interns with a formal offer letter and a completion certificate on successful conclusion of the program.

As part of its Web Development track, Prodigy InfoTech offered the author the position of Web Development Intern, providing the opportunity to work on real-world front-end projects, build practical coding skills, and contribute meaningfully to the team as part of a structured, educational internship program (Contact: contact@prodigyinfotech.dev | www.prodigyinfotech.dev).

3. Problem Statement

While foundational web development concepts such as HTML tags, CSS rules, and JavaScript syntax are commonly taught in academic coursework, translating that theoretical knowledge into fully functional, interactive, user-facing web components is a distinct and equally important skill. Students often lack exposure to building complete, self-contained front-end applications that combine structure, styling, and behaviour in a cohesive manner. The problem addressed by this internship was to design and build three independent, fully functional web components/applications — a responsive interactive navigation menu, a stopwatch application, and a two-player game — using only core HTML, CSS, and JavaScript, without relying on external frameworks or libraries.

4. Project Objectives

- To understand and apply semantic HTML for structuring web pages and UI components.
- To style fixed-position, responsive page elements using CSS.
- To implement scroll-based and hover-based interactivity using JavaScript event listeners.
- To use JavaScript timing functions to build accurate start, pause, and reset logic for a stopwatch.
- To track and display dynamic application state, such as lap times and game moves.
- To implement click-based event handling and win-condition logic for a two-player browser game.
- To gain hands-on, project-based experience in front-end web development within a structured internship environment.

4.5 Scope of the Project

The scope of the internship was limited to client-side, front-end web development using HTML, CSS, and JavaScript, without the use of any external frameworks, libraries, or back-end technologies. Three independent tasks were undertaken: (i) a responsive, fixed-position interactive navigation menu with scroll- and hover-based style changes; (ii) a fully functional stopwatch web application with start, pause, reset, and lap-time tracking; and (iii) a two-player Tic Tac Toe web application with click handling, game-state tracking, and win-condition checks. Each task was self-contained and did not involve server-side processing, databases, or deployment to a live production environment.

4.6 Technologies and Tools Used

- Markup: HTML5
- Styling: CSS3 (fixed positioning, responsive layout, hover- and scroll-based styling)
- Scripting: JavaScript (DOM manipulation, event listeners, timing functions)
- Browser Developer Tools, used for testing and debugging
- Code editor, used for writing and organising project files

4.7 System Architecture

Each of the three tasks completed during the internship followed a similar, layered client-side architecture:

- Structure Layer (HTML): Defines the semantic structure of the page or component — the navigation menu items, the stopwatch display, and the Tic Tac Toe game board.
- Presentation Layer (CSS): Controls the visual styling and layout, including fixed positioning of the navigation menu, styling of the stopwatch display for clear time reading, and structuring/styling of the game board.
- Behaviour Layer (JavaScript): Handles all interactivity and dynamic logic — detecting scroll and hover events for the navigation menu, implementing start/pause/reset and lap-time logic for the stopwatch, and handling click events, game-state tracking, and win-condition checks for Tic Tac Toe.

All three applications run entirely within the browser, with no server-side or backend component involved.

4.8 Methodology

The internship followed a task-based methodology, with each of the three assigned tasks building on the previous one in terms of complexity, as summarised below.

Task	Title	Description
Task 01	Responsive Loading Page – Interactive Navigation Menu	Created a fixed-position navigation menu, visible across all pages, that changes background colour and font colour on scroll and changes menu-item style on hover. HTML was used to structure the menu, CSS to style it and keep it fixed in position, and JavaScript to implement the scroll- and hover-based interactivity.
Task 02	Stopwatch Web Application	Built an interactive, user-friendly stopwatch that accurately measures and records time intervals. HTML was used to structure the interface, JavaScript functions were implemented to start, pause, and reset the stopwatch and to track and display lap times, and CSS was used to style the display for clear, accurate time reading.
Task 03	Tic Tac Toe Web Application	Built an interactive two-player Tic Tac Toe game in which players aim to get three markers in a row to win. HTML and CSS were used to structure and style the game board, while JavaScript functions handled user clicks, tracked each player's moves, and checked for a completed row, column, or diagonal to detect a win.

Each task was completed independently and reviewed as part of the internship's structured, educational project work.

4.9 Expected Outcomes

- Ability to structure web pages and UI components using semantic HTML.
- Ability to style fixed-position, responsive elements using CSS.
- Skill in handling scroll and hover events with JavaScript for interactive UI.
- Understanding of JavaScript timing functions to build accurate start/pause/reset logic.
- Ability to track and display dynamic state, such as lap times and game moves.
- Skill in implementing click-based event handling and win-condition logic for a game.
- A completed portfolio of three functional, browser-based web applications demonstrating core front-end development skills.

4.10 Certificate of Completion and Communication Proof

The Internship Offer Letter and the Certificate of Completion issued by Prodigy InfoTech, confirming successful completion of the 1-month Web Development internship with outstanding remarks, are attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

1. Prodigy InfoTech, Internship Offer Letter, Web Development Intern, dated 30th July 2026.

2. Prodigy InfoTech, Certificate of Completion, CIN: PIT/AUG26/00972, issued on 1st September 2026.

3. MDN Web Docs — HTML: https://developer.mozilla.org/en-US/docs/Web/HTML

4. MDN Web Docs — CSS: https://developer.mozilla.org/en-US/docs/Web/CSS

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6. W3Schools: https://www.w3schools.com/